

# Dissent Within a Transparent Central Bank: Causes and Effects \*

Jacob Granqvist<sup>†</sup>

First draft: 27 April 2026

Latest version: [October 6, 2026](#)

## Abstract

How does dissent feature in a central bank with fully transparent deliberations? By studying a uniquely transparent central bank, the Swedish Riksbank, I study in detail the causes and consequences of dissent in central bank communication. First, I map dissent to governor demographics and highlight some characteristics co-varying strongly with general dissent, for example, having a higher academic degree. Second, I show that a one-SD increase in general dissent reduces short term interest rate effects by roughly a third compared to the mean. Furthermore, studying the effects of the direction of dissent, I observe that dissenting in the dovish direction drives both shorter and longer rates upwards. Finally, when studying impulse responses from monetary policy shocks induced by transcript releases in different dissent regimes, I see evidence that dissent has an effect on the shape of the subsequent interest rate path. **Keywords:** *central bank communication; transparent central bank; dissent; monetary policy; text analysis; high frequency outcomes.*

## 1 Introduction

For a long time, the procedures of a central bank were something hidden and even mysterious to the public. The former chair of the Federal Reserve, Alan Greenspan, took pride in being able to, with his so called Fedspeak, perform his "mumbling with great incoherence". Since then, central banks have broadly adopted transparent communication frameworks. A transparent communication is believed to make the transmission of monetary policy more effective since it highlights the possible future economic scenarios, reflecting the underlying uncertainty about the future state of the world.

When contemplating dissent, it is vital to draw a line between dissent in actual extensive margin dissent (discrete decisions on voting and asset purchases) and intensive margin dissent (e.g. dissent in views on the economy, tendencies to prefer another rate). I posit that extensive margin dissent always implies intensive

---

\*I want to thank Martin Flodén, Klaus Adam and Tiago Bernadino for valuable comments and discussions. I also want to thank Edvin Ahlander, Per Krusell, Jinci Liu, Andreas Madestam, Hedda Thorell, Leiv Storesletten, David Strömberg, Lars E.O. Svensson and seminar attendants at the IIES Macro group and the Mannheim University Monetary Economics Students' Colloquium.

<sup>†</sup>IIES, Stockholm University, e-mail: [jacob.granqvist@iies.su.se](mailto:jacob.granqvist@iies.su.se)

margin dissent. However, the converse does not necessarily hold. There can be a lot of dissent in communication, without there being dissent in actual decisions. However, to study this type of dissent, one needs access to the deliberations leading up to each monetary policy decision, which is something that few central banks provide.

With that said, one central bank that is notably transparent is Sveriges Riksbank, the Swedish central bank. At the time of a monetary policy decision, the Riksbank sends out a press release the day after, containing the decision, a motivation and whether there was dissent<sup>1</sup>. Five to ten days later, the Riksbank publishes the transcripts of the meeting where the decision on the interest rate was made, which include an attributed transcript of each board member's contributions to the monetary-policy discussion. The timing of these releases are prespecified and expected by financial market. Using this uniquely transparent and structured nature of the Riksbank's monetary policy communication, one can answer vital questions concerning the effects of transparency in central bank communication.

The contributions of this paper are threefold. First, I develop a database that includes all statements from meetings of the executive board by individual governors from 2007 until today. This database consists also of voting records, as well as literal statements, but most importantly for my analysis, a sentiment score for each governor at each meeting. I build this sentiment score by employing an off-the-shelf LLM with which I map the statements of the governors in a hawkish/dovish dimension to get at each governor's monetary policy sentiment in each period. Using this sentiment variable, I create two new variables to measure dissent that captures dissent in communication, and enables me to get observe that can not be read off voting records.

Second, I use this database to examine if and how dissent in a central bank affects financial markets as well as real macro economic variables. When protocols are released, financial markets usually react to the content of these protocols. I document a novel fact that the size of financial market outcomes shrinks when there is more dissent on the board. The size of this effect is large and reduces the size of the shocks by about 33%. I also document that if there is a dovish dissenting minority, the more it deviates in the dovishness direction drives index swap rates upward.

Third, I also perform a dissent state decomposition of protocol releases by the amount of dissent in the board and thereafter perform state-dependent Jordà projections, obtaining the result that more dissenting board yields a 25 bps lower interest rate path in the medium run.

*Related literature.* The literature on central bank communication is a large and growing one. (Blinder et al. (2008)) is a useful reference that surveys the earlier part of the literature. One of the earlier contributions quantifying the text contents of central bank transcripts is Apel and Blix Grimaldi (2014). Using a dictionary approach, they create an index of aggregate board hawkishness over time, and show that this can predict future repo rate decisions, a feature also of the index developed in this paper. More recently, Apel et al. (2022) uses the same dictionary approach as the paper by Apel and Blix Grimaldi, but also use advances in textual analysis to create similarity measure to assess committee agreement. There they show that unity is higher

---

<sup>1</sup>Before 2009 dissent was not disclosed until the release of the transcripts. This reform seemed to, among other things, have a discipline effect on governors: Apel et al., 2026. Similar effect was observed in Hansen et al., 2018.

preceding periods of monetary tightening, showing support for distaste to raise interest rates until there is internal consensus.

The advent of large language models has created new ways of analysing textual information in general and central bank communication specifically. A study from the ECB (Geiger et al. (2025)) develops a Hawk-O-Meter (an LLM derived measure of the ECB executive board) to study the policy attitude of the ECB over time. Using this index, they provide time series of hawkishness over time in the ECB. Another recent work on this can be seen in Howes et al. (2026) and Baird et al. (2026). Howes et al. (2026) construct a data set of FOMC meeting transcripts to study governor specific monetary policy preferences. They show, among other things, that dissent can erode governor influence on future policy. However, due to institutional restrictions, they do not investigate the effect of dissent on market interest rates, which is something I contribute with in this paper.

Modeling central bank communication features in papers like Meade and Stasavage (2008), Riboni and Ruge-Murcia (2014), Blot et al. (2025) with models trying to rationalize among other things, dissenting behaviour within a central bank board. In Meade and Stasavage (2008), making transcripts public seems to have made FOMC governors less dissenting, in line with model predictions highlighting the risk of showing incompetence by dissenting and being incorrect. Career concerns then motivate sticking to the herd. Riboni and Ruge-Murcia (2014) studies individual governor vote data, obtained from (among other central banks) the Riksbank. However, the paper does not study transcripts and communicative dissent. They show results and model predictions, that dissent votes from certain governors can explain future changes in CB policy. Blot et al. (2025) also make use of dissent in voting, in the ECB as exposed in the monetary speeches given by the chair. On their paper they show that dissent affects stock prices negatively.

Layout of the paper: section 2 presents the data set, section 3 describes the method and some descriptives on the data set, section 4 presents the results and section 5 concludes.

## 2 Data

I have compiled an extensive data set on the communication and decision making of the Riksbank's executive board (EB). This has been possible due to the transparent nature of the Riksbank. The data set stands on three legs: EB meeting transcripts, press releases communicating EB decisions and voting records on monetary policy of the EB. Furthermore, I also have a range of financial market outcomes as macroeconomic outcomes. Details on the text extraction can be found in Appendix A.1.

### 2.1 Transcripts

The written transcripts from the meetings of the EB of the Riksbank have been collected from the Riksbank's website as well as from the Riksbank's archive website. These transcripts range from 1999-01-04 until 2025-09-01. However, in the database I include only transcripts from 2003-02-06 and onwards. The reason for this is that the transcripts from that date and onwards are structured in a similar fashion. There has been one major change in the composition of these transcripts over time. From 2007-06-19 the board decided to attribute

the statements<sup>2</sup> in the transcripts, making it possible to see exactly who said what during the meetings. This structure has remained until today. In total this gives us 140 transcripts, out of which 108 are transparent. A unique feature of the Riksbank, is that the transcripts are released at fixed day and time after each meeting of the EB. The timing is decided upon at each meeting of the board and is usually set to 09:30 one or two weeks after the meeting. A definitional note on the transcripts: I will throughout this paper use the term "transcript" over "minute". This is because of the nature of the monetary policy communication in the Riksbank. These texts are not summaries or reductions of what was said during the meetings (as a minute would be). They are (as much as possible) the literal transcription of what was said during each meeting of the EB.

## 2.2 Press releases

From the same web sources as above, I have also collected all press releases (PR) following a meeting of the EB. The PRs are presented to the public the following work day after the EB convened. They often contain a condensed message on the state of the economy as well as the decision on the interest rate. If there has been other decisions relating to quantitative easing, these are also communicated here. The format of the PRs have, similarly to the transcripts, changed over time. However, these changes have been less structural and more varying in the informativeness dimension. Therefore, I have included PRs from 1999 until 2025 in the database. Following a decision by the EB in April 2009 (Sveriges Riksbank, [2009-04-02](#)), it was decided that the board should reveal internal voting dissent already in the PR. If there is dissent, a one sentence motivation is also included. After the release of the PR a black-out period commences that ends at the publication of the transcripts. In total this gives us 167 PRs (101 transparent). Similar to the transcripts, these press releases are also released at a fixed day and time, usually at 09:30 the day after the meeting of the EB has occurred. An important feature for identification is that the press releases are followed by a black-out period until the release of the transcript. During this period the governors are only allowed to communicate the majority decision of the board.

## 2.3 Vote data of the EB

Publicly available is also time series data of the votes cast by the individual EB members from 1999-01-04 until 2026-09-22. In total there have been 199 votes held until today. The reason why there is not a perfect overlap with the amount of PRs and the number of votes is because of a previously informal nature of the Riksbank's proceedings. In the early 2000s, votes were cast a bit more irregularly. From 2003-02-06 and onwards, i.e. from the date which is the start date of the transcript data there have been 145 votes. The remaining irregularity is due to extra changes of the interest rate during the Great Recession and the onset of the Covid-19 pandemic. A definitional note on the composition of the EB: the EB of the Riksbank consisted of 6 members until 2022, and since then of 5 members. One of them is the chair of the EB while the others serve as deputy governors. If there is a tie, the chair is the tie breaker.

---

<sup>2</sup>I will henceforth call the anonymized transcripts *opaque*.

## 2.4 Monetary policy event study data base

Both PRs and transcripts throughout the history of the data I am using have been released on fixed dates and times. This makes it possible to identify the instantaneous effect of the information conveyed in them. I am using the database built by Almerud et al., 2024, which was most recently updated in December 2025. The database contains percentage point changes in OIS rates for a series of different maturities (1 month to 10 years) in a tight window (-10 min to +20 min) around the communication events. It also contains data on reactions of the SEK/USD, SEK/EUR exchange rates as well as the OMX SPI Swedish stock market index.

## 3 Digitization and descriptives

### 3.1 Developing the sentiment score

To analyze the contents of the text material, I have used an off-the-shelf large language model from Meta. The specific model I have used for this project is the `llama-3.3-70b-versatile` (Grattafiori et al., 2024). This model offers an "industry-leading performance in text and image understanding" (Meta AI, 2025). In addition, it is also a model that has been used earlier in similar work classifying central bank statements (e.g. Geiger et al., 2025). I have used the LLM to analyze the transcripts and the decisions from the meetings of the executive board sentence by sentence (for details about the prompting procedure, see Appendix A.2.1). Each sentence is then labeled as being either hawkish, dovish or neutral. For detailed instructions on how the LLM was instructed to label, see Appendix A.2.2. We should view hawkishness and dovishness to be existing in the same dimension and therefore mutually exclusive, if a sentence contains a stance on monetary policy.

To increase accuracy in the LLM classification, I let the language model keep the previous sentence as context when analyzing every next sentence after the first one in each statement. Adding more context sentences does not improve the accuracy of the classification. I also create a simple BOW algorithm to compare the sentiment score with. The LLM approach outperforms the BOW by being able to extract nuance in the text that the BOW can not.<sup>3</sup> Details on the validation and accuracy of the LLM can be found in Appendix A.2.3.

One can of course question the use of LLMs given that they are by construction opaque. However, I hope that the very transparent setup I provide in this paper can convince the reader that by using an LLM I can work around problems related to context affected classification that are difficult to solve with a BOW or bi-gram/trigram approach. I also want to emphasize that what I do is not something radical and new when it comes to classification and has been used multiple times before and is being used frequently<sup>4</sup>

When possible<sup>5</sup>, I create a governor-meeting specific sentiment score variable,  $S_{gt}$ , which is defined as follows:

$$S_{gt} = \frac{\#hawk - \#dove}{\#hawk + \#dove}$$

---

<sup>3</sup>For example the BOW algorithm incorrectly identifies a period of low dissent as having high dissent.

<sup>4</sup>In Geiger et al., 2025 and Fiore et al., 2024 for example.

<sup>5</sup>From June 2007 and onwards this is possible.

The sentiment score,  $S_{gt}$ , can be viewed as the net hawkishness of a governor  $g$  in time  $t$  and will naturally assume a value between  $-1$  and  $1$ <sup>6</sup>. If all sentences are regarded as neutral, the score is set to 0. The measure is similar to the ones defined in Apel and Blix Grimaldi, 2012 and Geiger et al., 2025. PRs have also been analyzed sentence by sentence, yielding an aggregate sentiment score for each meeting. I define this variable as  $S_t^p$ , which will never be governor specific. For more descriptive information on the underlying data set, look to Appendix A.3.

### 3.2 Descriptives

In Figure 1 we can see the meeting mean sentiment score, defined as  $S_t := \sum_g \frac{1}{|G_t|} S_{gt}$  where  $|G_t|$  is the number of governors attending a meeting in time  $t$ . I plot the score for the transcripts and the PRs as well as the changes in the repo rate over time in the graph below (the red and blue bars). As can be seen, there is a high degree of co-variation between the indices and the rate, except during the ZLB period beginning in 2016. I also conclude that there is, as perhaps is not too surprising, a high degree of co-variation between the sentiment score of the transcripts and the PRs ( $\rho \approx 0.87$ ). This is to be expected, since the PRs should summarize the contents of the transcripts.

---

<sup>6</sup>Thus, by construction, if a governor speaks a lot, but utters equally many dovish as hawkish sentences, she will be given a near 0 sentiment score for that meeting.

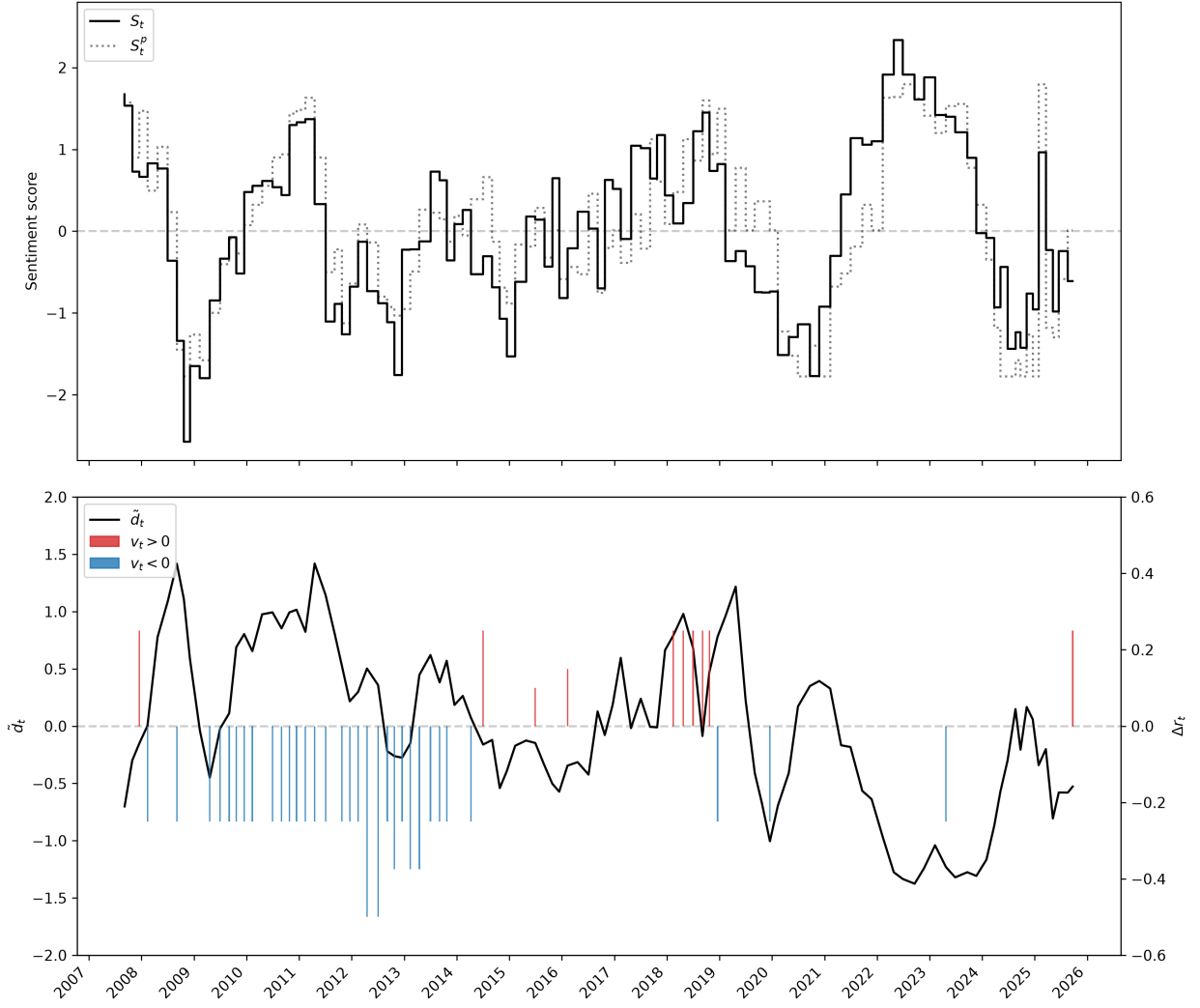


Figure 1: Mean sentiment score for transcripts and press releases over time (normalized) and repo changes, along with dissent (standardized) and vote dissent.

Since the transcripts of the Riksbank have been transparent since June 2007, I also inspect the trajectories of individual governor sentiment scores. One can thus also plot governor specific sentiment scores, which I do in Appendix A.4. Having data on individual level sentiment is what makes this data base unique and also what I will leverage in the next section.

### 3.3 A new measure of dissent

Formal dissent in the Riksbank can manifest itself in three ways. Either a member of the executive board disagrees on the interest rate, the path of the interest rate or asset purchases. These incidents of dissent are recorded in the voting transcripts of the Riksbank and binary in nature. Either you dissent on one of these three things, or you do not. However, dissent opinions in the board will not always, even seldom, materialize in actual dissent in these variables making the usage of votes only to capture dissent in a central bank quite coarse. Therefore, the new information on sentiment that can be obtained from the data set, makes it possible to understand nuances in dissent.

Using the governor specific sentiment score,  $S_{gt}$ , I define a more fine grained measure of dissent; the dispersion in terms of our sentiment score. I define governor dissent,  $d_{gt}$ , as follows:

$$d_{gt} := \left| S_{gt} - \sum_{i \neq g} \frac{S_{it}}{|G_t| - 1} \right| \quad (1)$$

which is a governor specific leave-one-out mean deviation variable. This gives us a way of measuring how far the governors are from each other in the sentiment score dimension. I furthermore define sentiment dispersion as

$$\tilde{d}_t := \frac{1}{|G_t|} \sum_g d_{gt} \quad (2)$$

where  $|G_t|$  is the number of governors voting at meeting in time  $t$ . Here, I use absolute dissent so that a meeting where there is dissent in both directions is not incorrectly misspecified as a low dissent meeting. In the lower graph of Figure 1, we plot the sentiment dispersion and overlay this with the average dissenting vote<sup>7</sup> on the interest rate at each meeting. As we can see, periods of dissent in voting of the interest rate often coincide with periods of high absolute meeting dispersion. However, there are also periods where  $\tilde{d}_t$  is high but it does not materialize in voting dissent. Therefore, I argue that this measure captures dissent that earlier was unobserved.

Next, I define a vector valued directional dissent variable. With the help of Defs. (1) and (2) I define what I call the *latent minority*,  $M_t$ :

$$M_t := \{g_t \in G_t : d_{gt} \geq \tilde{d}_t + \sigma_{d_t}/2\} \quad (3)$$

With the latent minority, I want to cluster the distant voices. Using this definition, we define the directional dissent variable  $D_{gt}$ :

$$\delta_{gt} := \begin{cases} S_{gt} - \frac{1}{|G_t| - |M_t|} \sum_{i_t \notin M_t} S_{it}, & g_t \in M_t \\ 0, & \text{else} \end{cases} \quad (4)$$

and then for the entire latent minority as a whole

$$\Delta_t := \begin{cases} \frac{1}{|M_t|} \sum_{g_t \in M_t} D_{gt}, & M_t \neq \emptyset \\ 0, & M_t = \emptyset \end{cases} \quad (5)$$

One might worry that the latent minority might be situated on both sides of the majority. This is a reasonable worry, but I can assure the reader that dissent in the Riksbank only occurs in one direction from the deciding minority. Therefore, I propose the above simple definition of a latent minority. Let us see how the latent mi-

---

<sup>7</sup>Definition of  $v_t$ , direction of dissent: the dissenting minority mean vote - the majority decision

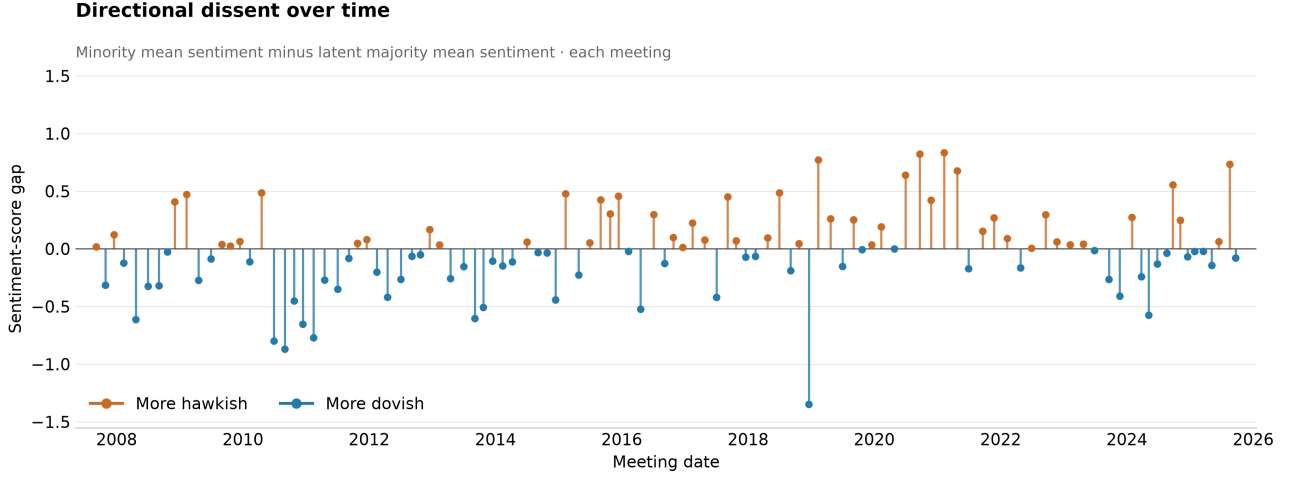


Figure 2: Directional dissent,  $\Delta_t$ , over time

nority directional dissent develops over time. We see that periods of high dovish dissent (2010-2012) and high hawkish dissent (2020-2022) are correctly identified in the data. In Appendix A.5, I also plot some governors and their directional dissent  $\delta_{gt}$  over time. In Appendix A.7, I elaborate on some another measure of dissent that are connected to voting dissent.

## 4 Results

### 4.1 Governor demographics

First, as a display of the great detail observed in the data, I correlate some governor characteristics, such as educational background and gender, with the dissent metric  $d_{gt}$ . I run a regression of dissent on all these variables as follows

$$d_{gt} = \alpha + \sum_i \beta_i \mathbb{1}\{\text{characteristic}_{ig}\} + \gamma \text{age}_{gt} + \delta \text{tenure}_{gt} + \text{year} + \epsilon_{gt}$$

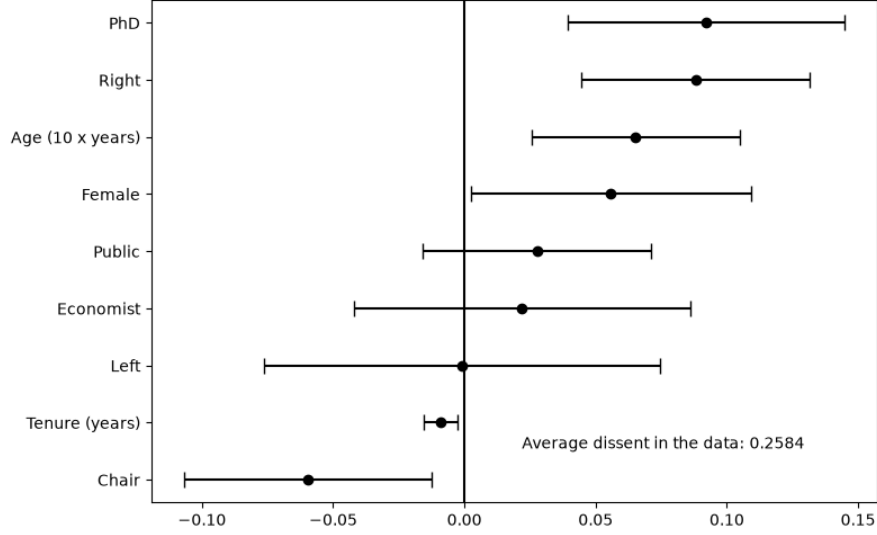


Figure 3: Absolute, standardized governor dissent correlated with governor demographics coefficient values with 95% confidence spans

The chair of the central bank is almost mechanically less dissenting, almost always being part<sup>8</sup> of the deciding majority in the sample I study. By years of tenure we observe less dissent, while age as well as having a PhD seem to correlate with more dissent.<sup>9</sup> The results on dissent and governor demographics could be viewed from the lens of Meade and Stasavage, 2008. Career concerns could be the explanation to why older people who have a safe academic position tend to dissent more. We also see support for the moderating role of the chair, as in Howes et al., 2026, where we see that the chair is significantly less dissenting than other members of the board. In Appendix A.6, I show the same results but using the directional dissent  $\delta_{gt}$  as the independent variable instead.

## 4.2 Interest rates

First, I establish a correlation between the sentiment score extracted from the transcripts and the changes in the interest rate. I regress the changes in the interest rate,  $\Delta r_t$ , on the mean of  $S_{gt}$ <sup>10</sup>. Furthermore, I also extend the regression to include  $S_t^p$ , which is the sentiment score of the related PR as well as the variable for dissent in the current and previous period and the interest rate in the previous period. I then also include controls,  $X_t$ , for inflation, unemployment and time fixed effects (year).

$$\Delta r_t = \alpha_0 + \beta_0 S_t + \beta_1 S_t^p + \beta_2 \Delta_t + \Delta r_{t-1} + \Gamma X_t + \varepsilon_t \quad (6)$$

As can be seen in Table 1, the sentiment score variable does correlate strongly with the repo rate set by the Riksbank. Controlling for the information content in the press release preceding the release of the transcripts, we see that the correlation of the sentiment score with the interest rate goes down, owing to some part to the fact that the interest rate decision already has been communicated in the PR. In Appendix A.8.1, I run the same

<sup>8</sup>The chair sides with the minority once in July 2014.

<sup>9</sup>These correlations hold also when dropping Lars E.O. Svensson, a more senior and very frequent dissenter in the data set.

<sup>10</sup>After taking the mean on meeting level, I standardize the  $S_{gt}$ .

regression with different weights for different members of the board, yielding no substantial differences.

Table 1: Results from regression (6)

|                       | (1)                 | (2)                 | (3)                | (4)                 | (5)                 |
|-----------------------|---------------------|---------------------|--------------------|---------------------|---------------------|
|                       | $\Delta r_t$        | $\Delta r_t$        | $\Delta r_t$       | $\Delta r_t$        | $\Delta r_t$        |
| $S_t$                 | 0.214***<br>(0.054) | 0.195***<br>(0.052) | 0.158**<br>(0.071) | 0.155**<br>(0.077)  | 0.140**<br>(0.067)  |
| Inflation controls    |                     | 0.062**<br>(0.025)  | 0.059**<br>(0.024) | 0.086***<br>(0.026) | 0.046*<br>(0.026)   |
| Unemployment controls |                     | -0.031<br>(0.067)   | -0.020<br>(0.070)  | 0.046<br>(0.073)    | -0.022<br>(0.065)   |
| $S_t^p$               |                     |                     | 0.044<br>(0.037)   | 0.035<br>(0.042)    | 0.017<br>(0.044)    |
| $\Delta_t$            |                     |                     |                    | -0.147**<br>(0.062) | -0.114**<br>(0.056) |
| $\Delta r_{t-1}$      |                     |                     |                    |                     | 0.379***<br>(0.117) |
| Constant              | -0.008<br>(0.016)   | 0.107<br>(0.509)    | 0.025<br>(0.533)   | -0.522<br>(0.558)   | 0.079<br>(0.516)    |
| R-squared             | 0.606               | 0.631               | 0.634              | 0.654               | 0.707               |
| F-stat                | 15.515              | 7.440               | 7.761              | 6.210               | 7.606               |
| No. obs.              | 140                 | 140                 | 140                | 113                 | 113                 |

Standard errors in parentheses

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

One important take away from this regression is the result in column (6). It is perhaps more of a sanity check, but it reinforces the fact that we are actually identifying a relevant latent minority. When there is hawkish deviation, i.e.  $\Delta_t > 0$ , this is correlated with lower interest rate changes, indicating that the hawkish dissent actually also materializes in hawkish dissenting vote outcomes (i.e. interest rate decreases) and vice versa.

### 4.3 Financial market outcomes

To estimate the causal effect of board dissent on financial markets, I exploit high-frequency changes in financial market rates within a tight window around the release of transcripts. These are changes to financial market rates that occur in a -10 min to +30 minute window around the release of the transcripts. I proceed by looking at both the size and direction of market outcomes,  $Y_t$ , as a function of the two dissent measures.  $Y_t$  is in this case long and short term interest rate swaps, as well as exchange rate reactions and stock exchange rate responses. I run multiple regressions, using both the dissent dispersion variable  $\tilde{d}$ , defined in (2) and also the directional dissent variable  $\Delta$ , defined in (5). It might be of interest to inspect the distributions of the

independent variables. These can be found in Appendix ?? . First I run the following two regressions:

$$Y_t = \alpha_0 + \beta_0 \Delta_t + \beta_1 S_t^{maj} + \beta_2 \text{vote}_{dissent} + \Gamma X_t + \varepsilon_t \quad (7)$$

$$Y_t = \alpha_0 + \beta_0 \tilde{d}_t + \beta_1 S_t^{maj} + \beta_2 \text{vote}_{dissent} + \Gamma X_t + \varepsilon_t \quad (8)$$

I include the majority sentiment  $S_t^{maj}$  as a control, as well as a dummy  $\text{vote}_{dissent}$  for if there was dissent. I also include controls  $X_t$ , including the level of the repo rate as well as a global VIX index as a proxy for economic risk. The VIX index is included to ensure that the results are not driven by the underlying aggregate risk state of the economy. Then I run the same regression but instead with the absolute size of  $|Y_t|$ . Results are reported in Table 3.

Let us first look at the results from regression (7). Looking at the vanilla regression, not restricting the independent variable in any way, we can not observe any statistically significant effects of directional dissent. However, if we narrow the scope and focus on dovish dissent, another picture emerges. In Table 2 we see that a dovish dissenting latent minority causes market rates to increase. This could appear odd at first glance. However, one possible explanation is that the market reads the dovish dissenters not getting their opinion through as an indication that market rates will become higher.

|                | (1)               | (2)                        | (3)                 | (4)                        | (5)                       | (6)                 | (7)               | (8)                      | (9)                      | (10)                |
|----------------|-------------------|----------------------------|---------------------|----------------------------|---------------------------|---------------------|-------------------|--------------------------|--------------------------|---------------------|
|                | 1M                | 3M                         | 6M                  | 1Y                         | 2Y                        | 5Y                  | 10Y               | SEK/USD                  | SEK/EUR                  | OMX                 |
| $\Delta_t < 0$ | 0.879<br>(1.620)  | <b>-4.577**</b><br>(1.831) | -0.914<br>(1.163)   | <b>-1.019**</b><br>(0.453) | <b>-0.967*</b><br>(0.546) | -0.619<br>(0.468)   | -0.593<br>(0.559) | <b>11.185</b><br>(7.861) | <b>12.699</b><br>(7.564) | 1.441<br>(8.433)    |
| Majority mean  | 1.916<br>(3.182)  | -2.671<br>(3.171)          | -0.911<br>(1.539)   | -1.063<br>(0.796)          | -1.284<br>(0.880)         | -0.407<br>(0.665)   | -0.450<br>(0.690) | 8.521<br>(11.545)        | 10.783<br>(10.230)       | -4.657<br>(19.503)  |
| Dissent votes  | -0.552<br>(0.803) | 0.048<br>(0.520)           | 0.208<br>(0.459)    | 0.502<br>(0.307)           | 0.137<br>(0.315)          | -0.061<br>(0.193)   | 0.011<br>(0.188)  | 3.251<br>(3.946)         | -0.004<br>(2.577)        | -12.932*<br>(7.164) |
| $r_t$          | 2.363<br>(1.866)  | 1.592<br>(1.342)           | -1.825**<br>(0.761) | -0.344<br>(0.453)          | -0.706<br>(0.557)         | -0.963**<br>(0.363) | -0.530<br>(0.368) | 10.103<br>(6.912)        | 10.860*<br>(6.074)       | 3.754<br>(13.156)   |
| VIX            | 0.930<br>(0.959)  | 0.114<br>(0.804)           | 0.282<br>(0.824)    | -0.677**<br>(0.285)        | 0.264<br>(0.442)          | 0.006<br>(0.292)    | 0.216<br>(0.280)  | 5.120<br>(3.869)         | 6.032*<br>(3.253)        | -8.602<br>(6.047)   |
| Constant       | -3.361<br>(2.571) | -2.649<br>(1.868)          | 2.868**<br>(1.074)  | -0.464<br>(0.574)          | 0.537<br>(0.754)          | 1.020*<br>(0.565)   | 0.425<br>(0.632)  | -16.936<br>(10.519)      | -14.923<br>(9.300)       | 1.310<br>(19.485)   |
| Year FE        | Yes               | Yes                        | Yes                 | Yes                        | Yes                       | Yes                 | Yes               | Yes                      | Yes                      | Yes                 |
| R-squared      | 0.333             | 0.534                      | 0.381               | 0.524                      | 0.312                     | 0.353               | 0.306             | 0.341                    | 0.276                    | 0.503               |
| Mean outcome   | -.145             | .425                       | .311                | -.37                       | -.437                     | -.407               | -.315             | .099                     | .354                     | -6.328              |
| F-stat         | 0.494             | 1.925                      | 1.234               | 3.807                      | 1.593                     | 2.566               | 2.496             | 0.949                    | 1.337                    | 1.280               |
| No. obs.       | 53                | 53                         | 53                  | 53                         | 53                        | 53                  | 53                | 53                       | 53                       | 53                  |

Standard errors in parentheses

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

Table 2: Studying the effect of negative directional dissent

The results in Table 2 are also robust to dropping the most frequent dissenter in the data set before calculating the average directional dissent (see Appendix A.11 for details).

Let us now continue by inspecting the results from the second regression, i.e. regression (8). We can observe these in Table 3.

|                   | (1)                       | (2)               | (3)                 | (4)                 | (5)                 | (6)                 | (7)                 | (8)                 | (9)                  | (10)                |
|-------------------|---------------------------|-------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|----------------------|---------------------|
|                   | 1M                        | 3M                | 6M                  | 1Y                  | 2Y                  | 5Y                  | 10Y                 | SEK/USD             | SEK/EUR              | OMX                 |
| $\tilde{d}_t$     | <b>-0.390*</b><br>(0.198) | -0.175<br>(0.407) | 0.081<br>(0.109)    | 0.095<br>(0.081)    | 0.124<br>(0.094)    | 0.059<br>(0.083)    | 0.064<br>(0.076)    | 0.433<br>(0.916)    | 0.423<br>(0.830)     | 0.594<br>(1.595)    |
| Majority mean     | -0.183<br>(0.656)         | -0.639<br>(0.816) | -0.417<br>(0.358)   | -0.131<br>(0.265)   | 0.161<br>(0.272)    | 0.183<br>(0.222)    | 0.239<br>(0.253)    | -2.610<br>(3.580)   | -2.979<br>(2.916)    | -0.742<br>(5.956)   |
| Dissent votes     | 0.268<br>(0.405)          | 0.067<br>(0.294)  | -0.124<br>(0.181)   | -0.360*<br>(0.183)  | -0.184<br>(0.151)   | -0.107<br>(0.118)   | -0.145*<br>(0.083)  | -0.730<br>(1.994)   | -1.908<br>(1.394)    | 2.796<br>(4.401)    |
| $r_t$             | -0.450<br>(0.747)         | 0.307<br>(0.587)  | -0.025<br>(0.371)   | 0.124<br>(0.235)    | -0.040<br>(0.238)   | -0.294<br>(0.226)   | -0.322<br>(0.286)   | 1.448<br>(2.758)    | -1.188<br>(1.803)    | -4.211<br>(5.667)   |
| VIX               | -0.344<br>(0.313)         | 0.017<br>(0.436)  | -0.101<br>(0.165)   | 0.170<br>(0.192)    | 0.023<br>(0.190)    | -0.150<br>(0.184)   | -0.089<br>(0.152)   | -1.813<br>(1.766)   | -2.360*<br>(1.361)   | 7.465**<br>(3.699)  |
| Constant          | 1.525**<br>(0.716)        | 0.987*<br>(0.576) | 1.170***<br>(0.406) | 0.652***<br>(0.235) | 0.754***<br>(0.282) | 1.046***<br>(0.295) | 1.091***<br>(0.337) | 9.447***<br>(3.531) | 11.611***<br>(2.568) | 17.990**<br>(7.489) |
| Year FE           | Yes                       | Yes               | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 | Yes                  | Yes                 |
| R-squared         | 0.367                     | 0.190             | 0.392               | 0.460               | 0.299               | 0.325               | 0.299               | 0.157               | 0.301                | 0.380               |
| Mean outcome size | 1.14                      | 1.281             | 1.007               | .559                | .621                | .677                | .668                | 10.42               | 8.694                | 14.976              |
| F-stat            | 1.049                     | 0.163             | 0.631               | 2.327               | 0.866               | 1.063               | 1.813               | 0.510               | 0.975                | 1.421               |
| No. obs.          | 107                       | 107               | 107                 | 107                 | 107                 | 107                 | 107                 | 107                 | 107                  | 107                 |

Standard errors in parentheses

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

Table 3: Results from regression (8), absolute

What can be observed is that when there is a large amount of dissent in the board, i.e. a high  $\tilde{d}_t$ , the size of the response on the short interest rates is substantially smaller than the average. A one standard deviation increase in  $\tilde{d}_t$  reduces the effect on the 1M interest rate by 33%. A similar effect exists also for the 3M interest rate, although smaller and non-significant. This result remains when put through a number of robustness tests (see Appendix A.10) such as dropping the most dissenting governor and dropping the 5% highest and lowest outcomes of the 1M variable. It also holds if we drop the meetings when there was dissent in voting. Importantly, using dissent in voting instead as the independent variable, yields no statistically significant effect on financial market outcomes.

What is it that drives this effect? Is it the uncertainty around the releases of the transcripts, induced by the dissent on the board, that reduces the ability of the CB to guide the market in a given direction? Due to the small amount of trade that occurs in these short term swaps, it is difficult to get numbers on volatility in the event windows. Nevertheless, I did run regressions on two different measures of volatility and did not find any significant results. See Appendix A.9 for details. Thus, this explanation can tentatively be ruled out. Still, more dispersion, as measured by  $\tilde{d}_t$  seems to dampen the short-term policy response. However, the most important effect to observe here might be the 0 effect for all other financial market outcomes. A more dissenting board does not seem to make the size of financial market outcomes larger. This lends support to the Riksbank's and other central banks' decision to increase transparency in communication. There is at least nothing that overtly showing dissent has an adverse effect on markets, except for in the case of the short run.

#### 4.4 State dependent impulse response functions

The results in Section 4.3 call for an examination if there are persistent effects at longer horizons. To extend the analysis of the economic consequences of dissent, I perform an analysis in the vein of Jordà, 2005. I use

normalized shocks to the 1 month overnight index swaps directly in the state dependent Jordà local projection as follows:

$$y_{t+h} - y_{t-1} = \alpha_h + \kappa_h \mathbb{1}\{\tilde{d}_t \geq \eta\} + \beta_h^H \mathbb{1}\{\tilde{d}_t \geq \eta\} \hat{i}_t + \beta_h^L (1 - \mathbb{1}\{\tilde{d}_t \geq \eta\}) \hat{i}_t + \sum_{l=1}^{\ell} \gamma_h X_{t-l} + \varepsilon_{t+h}, \quad h = 0, \dots, 36,$$

The details of this regression are the following: The outcome variables,  $y$ , are in this case CPIFxe inflation rate, repo rate, unemployment and an index for industrial production (IPI).  $\eta$  is the median of the dissent variable.  $X_{t-l}$  contains the controls which include 12 lags to the outcome variables and 12 lags to the shock variable. The data set has been divided into two states, either there is high dissent or low dissent, where the splitting line is the median of dissent in the data set. I then estimate  $\beta_h$  and  $\delta_h$  the state dependent impulse response at horizon  $h$  to a 1 SD shock to the 1M OIS swap rate for all horizons from  $h = 0$  to  $h = 36$ . For the CPIF, unemployment and IPI the responses are not different by state and insignificant on most time horizons. However, we see that the effects on the repo rate path is significantly different between states and that in the case of a low dissent regime results in a 25 bps higher interest rate path, all else equal. There could be multiple explanations to this result. On the one hand, if we see dissent as an outcome and not as a cause, a board might be experiencing high dissent right at the end of interest increasing/decreasing cycles. On the other hand, if dissent is seen as instrumental in steering policy, we can view it as a way that actually signals to the market that there will be no more interest rate hikes and therefore adjusts the path.

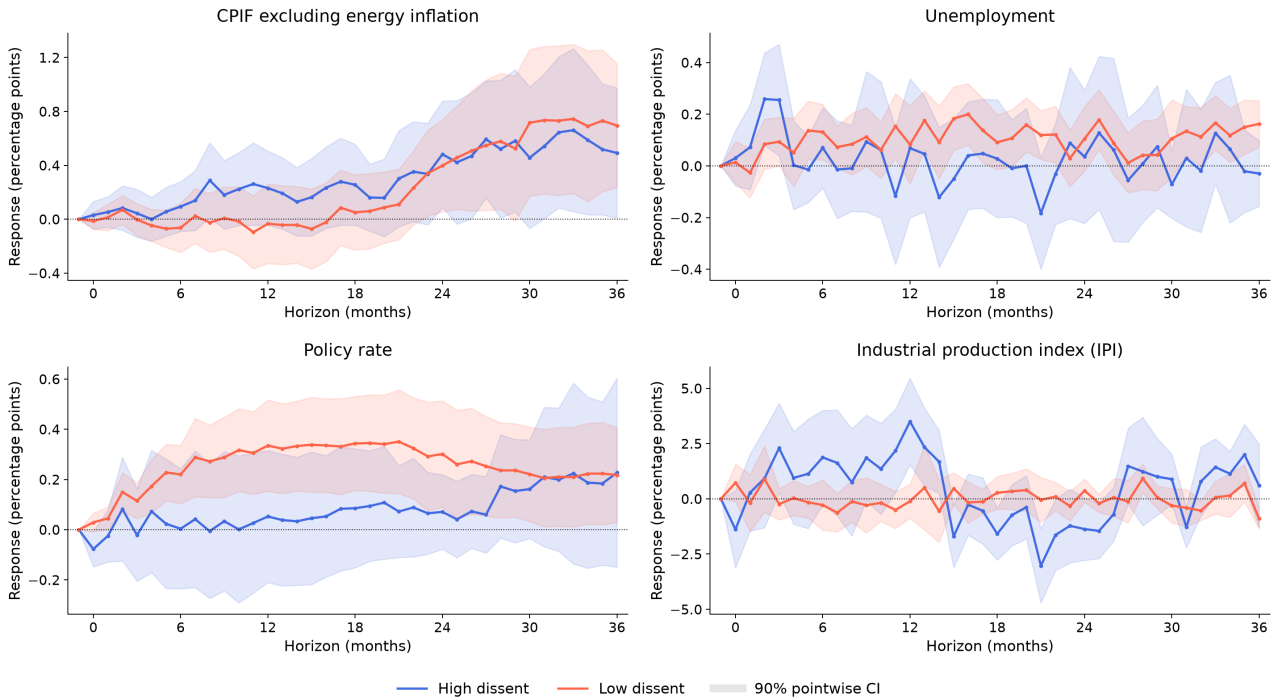


Figure 4: Impulse response functions of 4 macroeconomic variables to shocks to a 1M OIS swap rates by dissent regime.

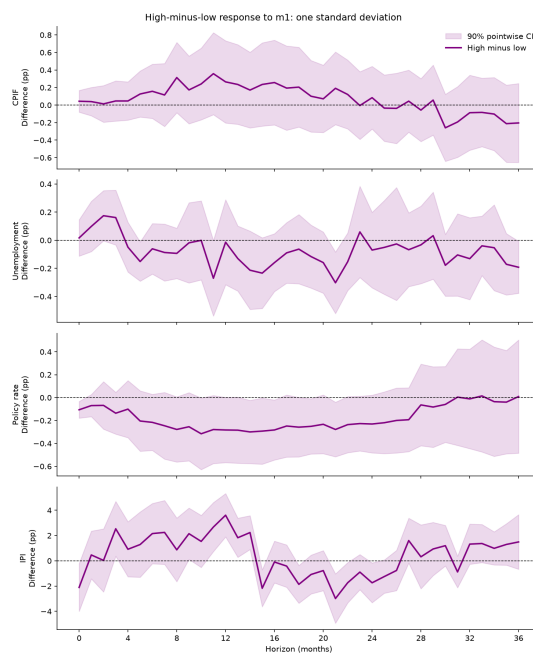


Figure 5: Difference between coefficient estimates by time horizon

## 5 Conclusion

In this paper I provide a novel and unique data set on central bank communication, employing data from a uniquely transparent central bank, the Riksbank, to study the effects of communicative dissent in transparent central bank. Using an off-the-shelf LLM to analyze the transcripts from the meetings of the executive board, I am able to map governor specific sentiment (hawk/dove) and in extension the dissent between governors on this topic over time. Then, using the dataset, I proceed to quantify the short run financial market effects of dissent in central bank communication. At the release of the transcripts, short-term financial market swap rate responses are dampened by around 33% when there is high dissent. Furthermore, when there is dovish dissent in the board, markets expectations of interest rates increase slightly, but significantly. Furthermore, looking at impulse response functions during high and low dissent regimes, we see that they are virtually the same for most outcome variables, while showing diverging and significantly different interest rate paths, lending support to dissent being a monetary policy regime indicator. In conclusion, the insight that communicative dissent in a transparent central bank has some minor effects in the short run effects, but as it seems, none or small effects in the longer run suggests that a central bank conducting transparent monetary policy does not induce unwanted volatility in the markets that affects the economy in an undesired way.

# A Appendix

## A.1 Details on parsing and data extraction

To extract text data from the transcripts, I use pdfplumber. The transcripts are divided into different sections with clearly (albeit changing) defined headers that make relatively easy to use regular expressions to extract the statements. The typical transcript is divided into four or some times five sections where I focus on the parts titled "Discussion" or "Economic outlook and monetary policy" which capture the deliberations of the governors. I assemble each governor statement as the collection of all governor utterances in a meeting of the executive board. I add interjection at the end of each governor statement to capture all the utterances of a governor at each meeting.

## A.2 LLM and classification

### A.2.1 Prompts

Model used: llama-3.3-70b-versatile

System prompt: You are a macroeconomist specializing in monetary policy and central bank communication analysis. You classify central bank statements at the sentence level.

Instruction:

Analyse the TARGET SENTENCE using the PREVIOUS SENTENCE as context (if provided). Also assign a probability - the confidence that the assigned score is correct. For each category, assign the required integer score and provide a short justification. If no previous sentence is provided, classify based on the target sentence alone. If unclear, assign score 0 and probability 0.6. Keep motivations concise (maximum 2 sentences). Do not restate either the target or the previous sentence. Base the classification strictly on explicit content. Do not infer policy intentions unless clearly implied. Return only valid JSON. Do not include any text before or after the JSON.

Definitions:

1. Economic Sentiment:

-1 = Negative economic outlook

0 = Neutral

1 = Positive economic outlook

2. Emotion:

0 = Objective, technical, neutral tone

1 = Emotionally charged or expressive language

3. Uncertainty:

0 = No explicit uncertainty expressed

1 = Explicit uncertainty about economic conditions or outlook

4. Hawkishness:

1 = Hawkish (if sentence provides information that suggests an increased likelihood of monetary tightening or implicitly implies such a stance. This includes statements on increasing inflation or inflation pressure, increases in purchasing power or real incomes that signal excess demand or overheating risk, adverse supply shocks, low unemployment and higher wage growth, accelerating economic activity and a strong or even unsustainable money or credit growth. Furthermore, statements on public debt (a reduction in the deficit or avoidance of public debt is considered hawkish) or support for inflation-reducing economic policies should be classified as hawkish.)

-1 = Dovish (if a sentence provides information that suggests an increased likelihood of monetary easing or implicitly implies such a stance. This includes statements on decreasing inflation or easing inflation pressure, decreases in purchasing power or real incomes that signal weak demand or slack in the economy, favorable supply shocks, rising unemployment and lower wage growth, decelerating economic activity and weak or contracting money or credit growth. Statements that explicitly prioritize or express preference for supporting economic growth, employment, or output over price stability should also be classified as dovish. Furthermore, statements on public debt (an increase in the deficit or acceptance of higher public debt is considered dovish) or support for debt-financed government spending and other inflation-tolerant or growth-stimulating economic policies should be classified as dovish.)

0 = Neutral (descriptive statements without economic reference or statements on economic conditions without implications for monetary policy)

Output strictly valid JSON: {"sentiment":

```
{ "motivation": string, "score":int },
```

"emotion":

```
{ "motivation": string, "score":int },
```

"uncertainty":

```
{ "motivation": string, "score":int },
```

"hawkishness":

```
{ "motivation": string, "score":int } }
```

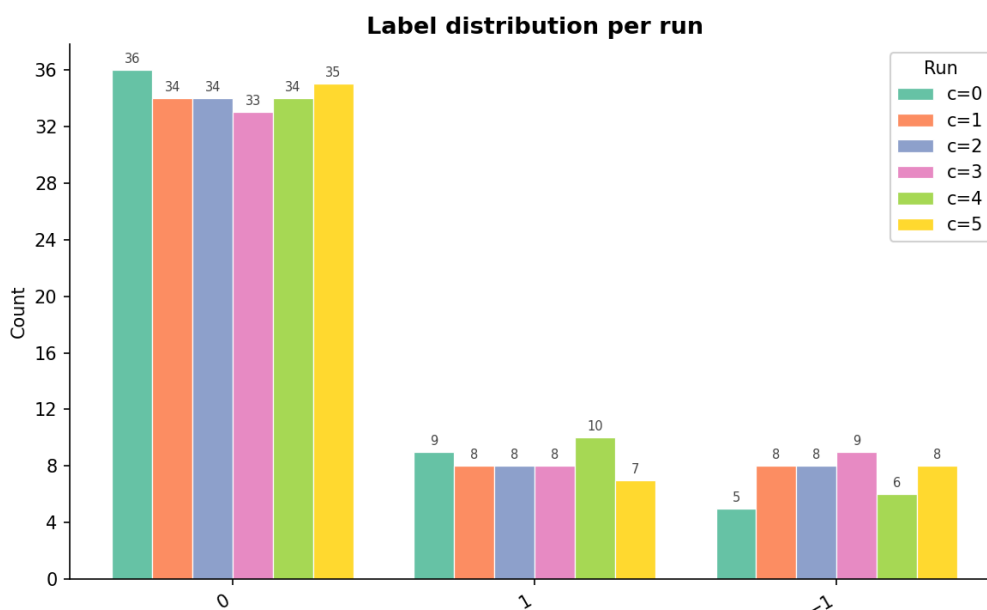
## A.2.2 Details on labelling

| A sentence is defined as... | ...if it...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>hawkish</b>              | <ul style="list-style-type: none"> <li>• indicates increased likelihood of monetary tightening or mentions rising inflation or inflationary pressures</li> <li>• references low unemployment, accelerating wage growth or signals excess demand</li> <li>• describes accelerating economic activity, unsustainable money growth or notes adverse supply shocks</li> <li>• encourages deficit reduction, public debt avoidance or in some way endorses inflation-reducing economic policies.</li> </ul> |
| <b>dovish</b>               | <ul style="list-style-type: none"> <li>• indicates likelihood of monetary loosening or mentions falling inflation or declining inflationary pressure</li> <li>• references concern about rising unemployment or describes weak or slowing economic activity</li> <li>• signals insufficient demand and supports debt-financed government spending.</li> </ul>                                                                                                                                          |
| <b>neutral</b>              | <ul style="list-style-type: none"> <li>• gives descriptive statements without economic reference or economic statements without implications for monetary policy.</li> </ul>                                                                                                                                                                                                                                                                                                                           |
|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

Table 4: Details on sentence classification

## A.2.3 Validation

Here I will show some results on how the LLM classification overlaps, depending on the number of context sentences. First, I classify the same 50 sentences with between 0 and 5 sentences of context to study the performance of the LLM.



I also perform a simple BOW measure of hawkishness to compare the performance of the LLM to this more

simple measure. I plot the analog to Figure 1 but also including the indices using a bag of words using the following two bags:

- **Hawkish words:** inflation, kpif, konsumentpris, producentpris, energipris, oljepris, råvarupris, utbudsstörning, kron, växelkurs
- **Dovish words:** tillväxt, resursutnyttjande, sysselsättning, konjunktur, finansiella, bnp, skuldsättning, bostadsmarknad, arbets, real

I then calculate a relative count score as

$$R_{gt} = \frac{\#hawk - \#dove}{\#hawk + \#dove}$$

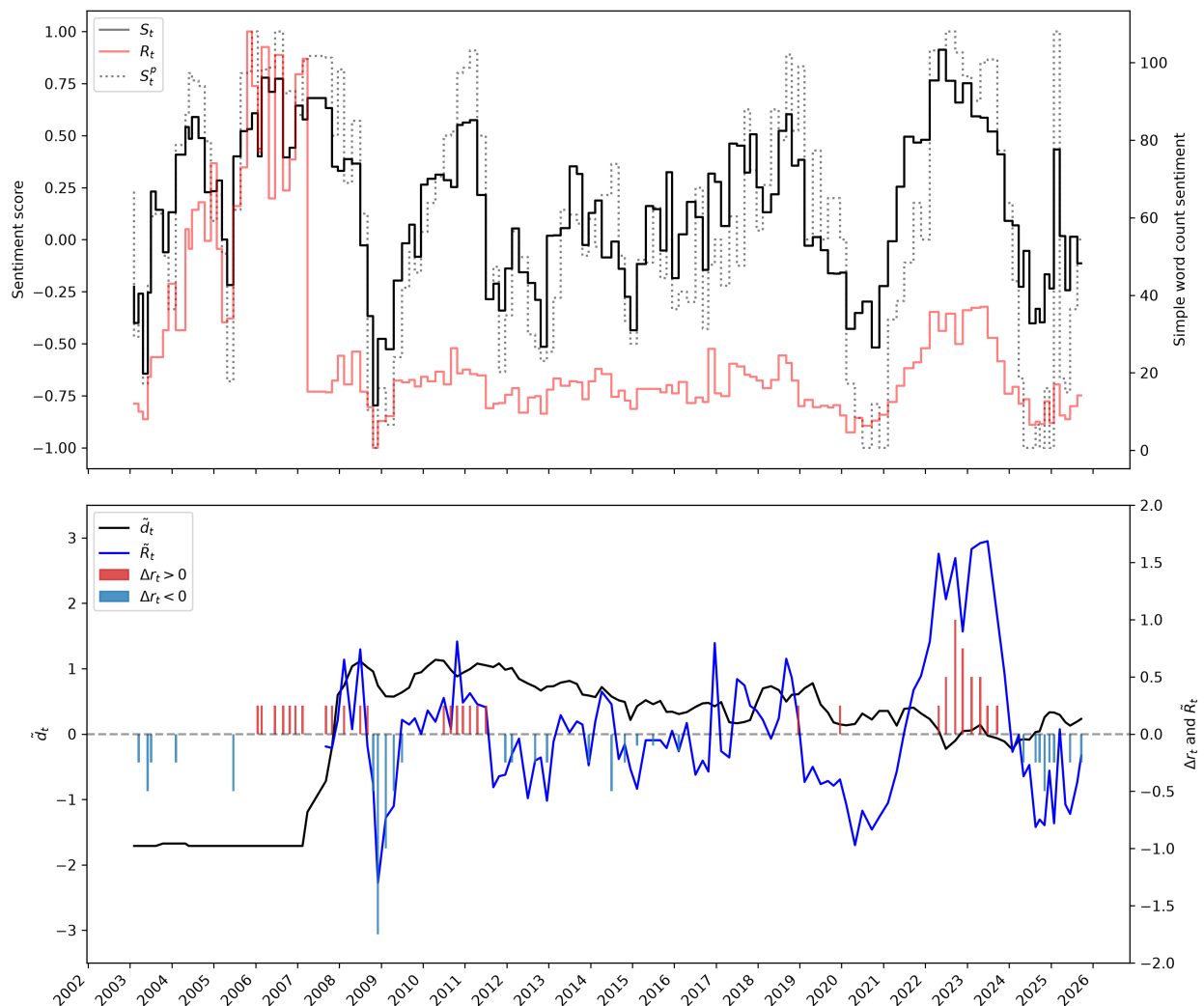


Figure 6: A reproduction of Figure 1 but also with simple BOW-index

What I see in Figure 6 is that the simple BOW algorithm yields little variation over time, and wrongly identifies the period in 2022-2024 as a period of high dissent. However, for the sentiment score I see only a tiny level of covariation and our  $\rho \approx 0.15$ . The major macroeconomic events in 2006-2009 and 2021-2024 are

somewhat picked up also by the simple bow-index.

### A.3 Summary statistics

Table 5: Summary statistics collapsed by date

| (1)             |          |         |      |       |      |
|-----------------|----------|---------|------|-------|------|
|                 | No. obs. | Average | S.d. | Min   | Max  |
| $S_t$           | 140      | 0.00    | 1.00 | -2.71 | 2.15 |
| $S_t^p$         | 140      | 0.00    | 1.00 | -1.89 | 1.65 |
| $\tilde{D}_t$   | 115      | -0.02   | 0.35 | -1.35 | 0.83 |
| $\Delta r_t$    | 140      | -0.01   | 0.27 | -1.75 | 1.00 |
| $r_t$           | 140      | 1.39    | 1.50 | -0.50 | 4.75 |
| $v_t^{min}$     | 48       | -0.15   | 0.34 | -0.75 | 0.75 |
| $D_t$           | 48       | -0.92   | 1.47 | -3.00 | 3.00 |
| Governor absent | 3        | 1.00    | 0.00 | 1.00  | 1.00 |
| $\alpha_{gt}$   | 8        | 2.12    | 0.83 | 1.00  | 3.00 |
| $\iota_{gt}$    | 46       | 1.87    | 0.58 | 1.00  | 3.00 |

Table 6: Summary statistics by governor over time

| Governor             | Start   | End     | mean( $S_t$ ) | sd( $S_t$ ) | mean(length) | sd(length) | sum( $ D_{it} $ ) | No. votes |
|----------------------|---------|---------|---------------|-------------|--------------|------------|-------------------|-----------|
| Henry Ohlsson        | 2015-02 | 2023-02 | 0.317         | 0.500       | 62.391       | 19.470     | 7                 | 46        |
| Lars Nyberg          | 2007-09 | 2011-12 | 0.248         | 0.446       | 71.074       | 23.452     | 2                 | 27        |
| Svante Öberg         | 2007-09 | 2011-12 | 0.228         | 0.512       | 60.593       | 17.178     | 1                 | 27        |
| Stefan Ingves        | 2007-09 | 2022-11 | 0.213         | 0.422       | 76.852       | 24.292     | 1                 | 90        |
| Kerstin af Jochnick  | 2012-02 | 2019-09 | 0.188         | 0.337       | 54.156       | 12.708     | 1                 | 45        |
| Cecilia Skingsley    | 2013-07 | 2022-06 | 0.107         | 0.426       | 47.604       | 10.569     | 0                 | 53        |
| Anna Breman          | 2019-12 | 2025-09 | 0.100         | 0.473       | 67.743       | 14.912     | 2                 | 35        |
| Erik Thedéen         | 2023-02 | 2025-09 | 0.089         | 0.316       | 65.421       | 19.314     | 0                 | 19        |
| Per Jansson          | 2012-02 | 2025-09 | 0.089         | 0.398       | 68.852       | 17.317     | 2                 | 82        |
| Aino Bunge           | 2023-02 | 2025-09 | 0.059         | 0.459       | 54.684       | 11.392     | 0                 | 19        |
| Martin Flodén        | 2013-07 | 2024-05 | 0.037         | 0.369       | 59.651       | 16.086     | 7                 | 63        |
| Barbro Wickman-Parak | 2007-09 | 2013-04 | 0.022         | 0.512       | 83.800       | 30.629     | 1                 | 35        |
| Anna Seim            | 2024-06 | 2025-09 | -0.079        | 0.420       | 39.909       | 8.666      | 1                 | 11        |
| Irma Rosenberg       | 2007-09 | 2008-12 | -0.136        | 0.494       | 58.333       | 31.093     | 2                 | 9         |
| Karolina Ekholm      | 2009-04 | 2014-09 | -0.170        | 0.259       | 67.212       | 23.738     | 18                | 33        |
| Lars E.O. Svensson   | 2007-09 | 2013-04 | -0.202        | 0.427       | 124.118      | 67.728     | 23                | 34        |

Notes: The governor specific data set begins in 2007-09. Some governors have this as a start date because they were already part of the board when the data became transparent.

A.4 Governor specific sentiment

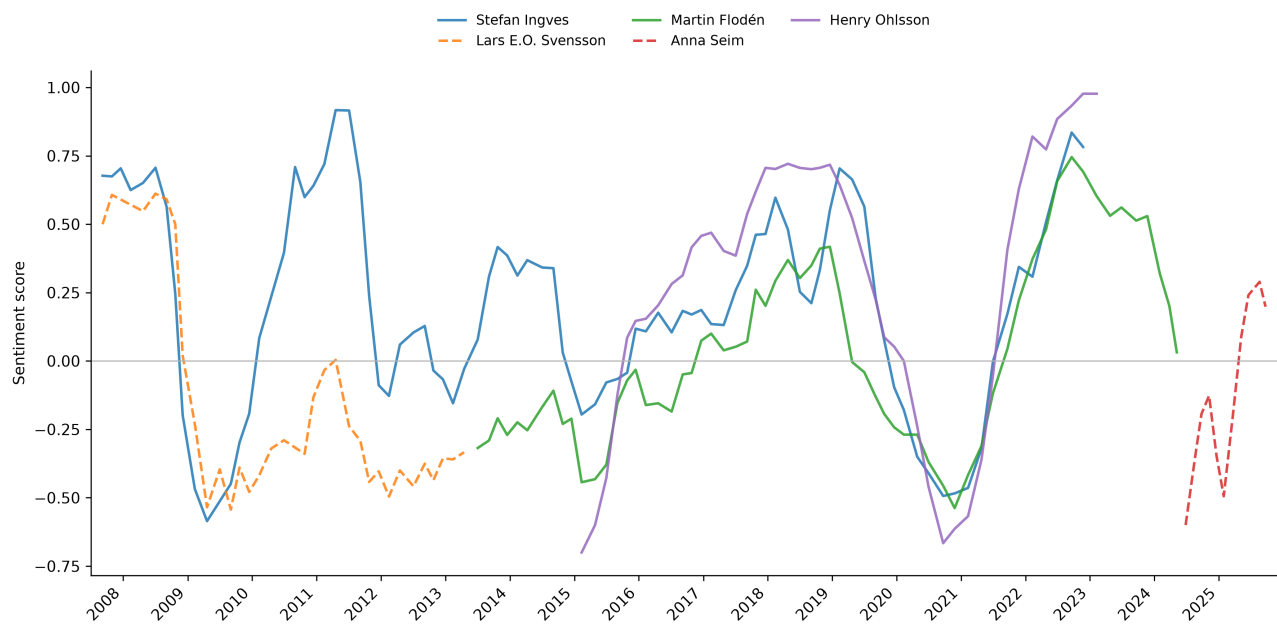


Figure 7: Governor specific sentiment scores over time.

## A.5 Governor specific directional dissent

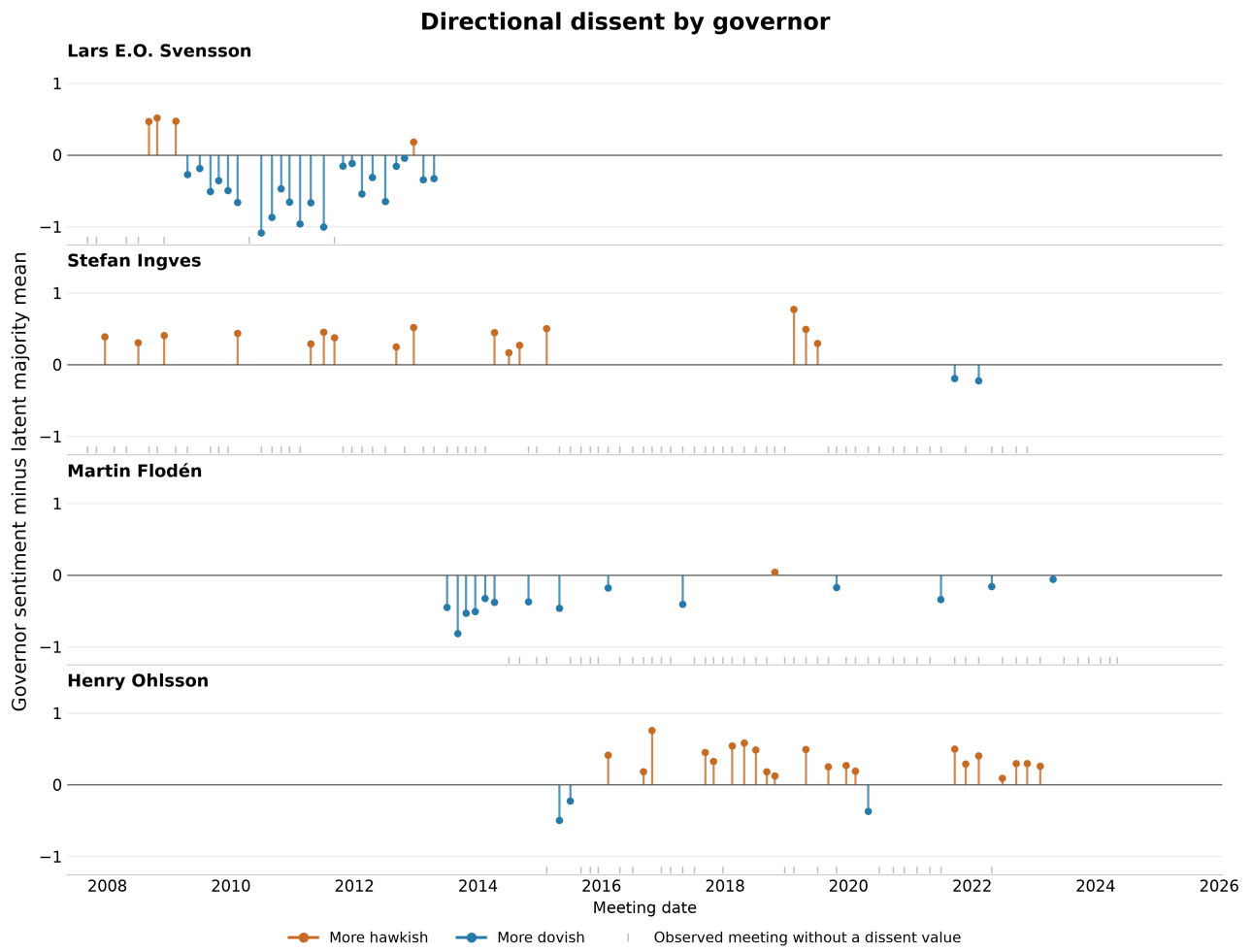


Figure 8: Governor, directional dissent over time.

## A.6 Governor characteristics and directional dissent

$$\delta_{gt} = \alpha + \sum_i \beta_i \mathbb{1}\{\text{characteristic}_{ig}\} + \gamma \text{age}_{gt} + \delta \text{tenure}_{gt} + \text{year} + \epsilon_{gt}$$

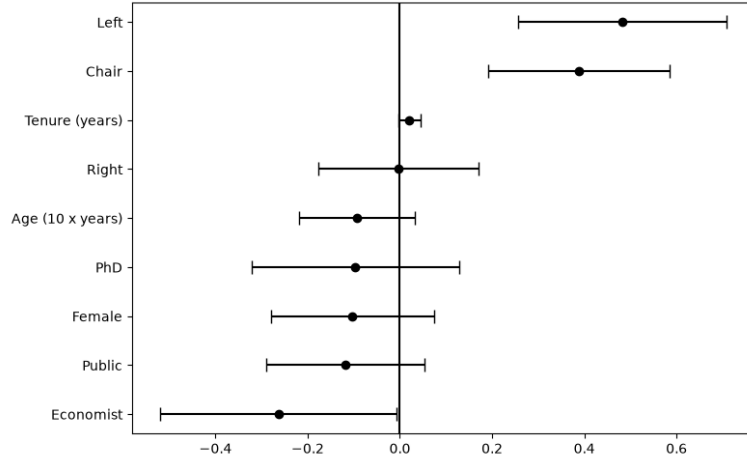


Figure 9: Governor directional dissent correlated with governor demographics coefficient values with 95% confidence spans

## A.7 Conventional measures of dissent

I define three variables that capture these three types of dissent:

Vote dissent:  $\nu_{gt} := v_{gt} - \Delta r_t$

Interest path dissent:  $\iota_{gt} \in \{0, 1\}$ , indicator whether governor  $g$  dissented to interest rate path at time  $t$

Asset purchases dissent:  $\alpha_{gt} \in \{0, 1\}$  indicator whether governor  $g$  dissented to asset purchases at time  $t$

where  $v_{gt}$  is the vote of governor  $g$  and  $\Delta r_t$  is the repo rate change decided by the board, all in time  $t$ .

Table 7: Correlation of different dissent variables

| (1)           |              |               |            |           |            |            |              |          |               |
|---------------|--------------|---------------|------------|-----------|------------|------------|--------------|----------|---------------|
|               | $\iota_{gt}$ | $\alpha_{gt}$ | $\nu_{gt}$ | $d_{gt}$  | $ D_{gt} $ | $ d_{gt} $ | $ \nu_{gt} $ | $S_{gt}$ | $\delta_{gt}$ |
| $\iota_{gt}$  | 1.000        |               |            |           |            |            |              |          |               |
| $\alpha_{gt}$ | -0.050       | 1.000         |            |           |            |            |              |          |               |
| $\nu_{gt}$    | -0.536***    | 0.028         | 1.000      |           |            |            |              |          |               |
| $d_{gt}$      | -0.260***    | 0.031         | 0.356***   | 1.000     |            |            |              |          |               |
| $ D_{gt} $    | 0.798***     | -0.048        | -0.583***  | -0.287*** | 1.000      |            |              |          |               |
| $ d_{gt} $    | 0.310***     | 0.061*        | -0.206***  | -0.027    | 0.230***   | 1.000      |              |          |               |
| $ \nu_{gt} $  | 0.779***     | -0.046        | -0.679***  | -0.290*** | 0.958***   | 0.224***   | 1.000        |          |               |
| $S_{gt}$      | -0.160***    | 0.012         | 0.255***   | 0.550***  | -0.171***  | -0.134***  | -0.181***    | 1.000    |               |
| $\delta_{gt}$ | -0.383***    | 0.067         | 0.453***   | 0.982***  | -0.387***  | -0.092     | -0.389***    | 0.664*** | 1.000         |

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

As we can see, dissenting on the interest path is strongly correlated to dissent in voting and also to dissent in sentiment. The negative sign on the correlations between  $\iota_{gt}$  and  $\nu_{gt}$  and  $d_{gt}$  stems from the fact that throughout the time series, dissent has occurred more often in the dovish direction ( $\sim 2007$ -2014). As for dissent in the asset purchasing dimension,  $\alpha_{gt}$ , we see that it correlates very weakly with  $|d_{gt}|$  and not with any of the other variables. A new measure of dissent that I study using the sentiment score is the mean difference in the sentiment score of the majority and the minority when there is vote dissent. We overlay this with the

mean difference in voting on the interest rate of the majority vs. the minority ( $\frac{1}{|G_t|} \sum_g \nu_{gt} : \nu_{gt} := v_{gt} - \Delta r_t$ ) and plot this in Figure 10 to highlight the way in which vote dissent and sentiment dissent covary.

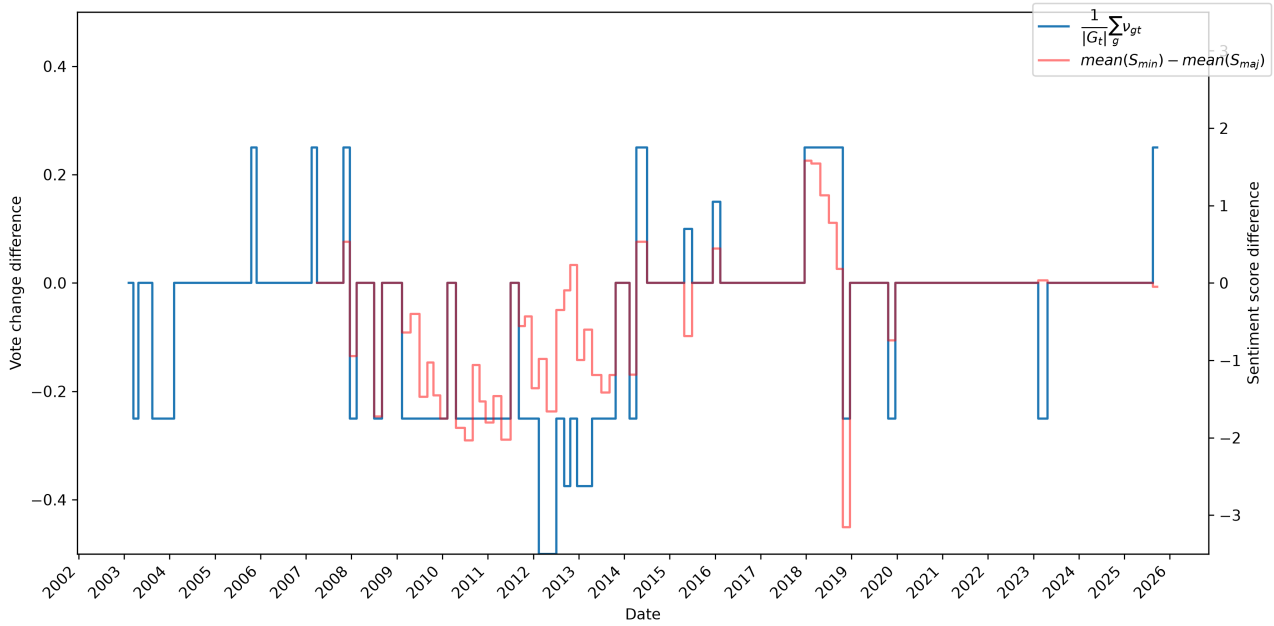


Figure 10: Intensity in voting dissent and sentiment deviation of dissenting minority

In the figure, we can see that there are two periods with consistent dissent. One dissenting dovish period beginning in 2009 where one to two members of the executive board consistently disagreed on policy and deviated, both in speech and in voting in a dovish direction. The peak dissent deviation with the majority of the board in this first period occurred in January of 2011. After that, there was some moderation in the deviation of the minority where eventually the prominent dissenters a bit later left the board. Following this, there is from 2018 and onwards a dissenting hawkish period which ends with the data set's biggest outlier in the dovishness dimension.<sup>11</sup> Before we have individual governor sentiment data, we also observe dissent but only in the voting. As I see, when data is available for the dissenting minority's sentiment, it tracks voting dissent quite well. There are only a few notable exceptions when dissent in voting is in the positive direction and sentiment deviates in the other direction, for example the 15 July 2015<sup>12</sup>.

To put the governor dissent variables into context, I provide some correlations below in Table 7.

<sup>11</sup>Governor Per Jansson expresses great concerns for too weak inflation and wants to keep the interest rate unchanged at -0.5 %.

<sup>12</sup>The governor in that case, Henry Ohlsson, still saw it as necessary to stimulate the economy but expressed a distaste for letting the interest rate get more negative.

## A.8 Alternate regressions

### A.8.1 Interest rate dependence with different weights

Table 8: Weighted by number of sentences spoken

|                       | (1)                  | (2)                 | (3)                | (4)                 | (5)                 |
|-----------------------|----------------------|---------------------|--------------------|---------------------|---------------------|
|                       | $\Delta r_t$         | $\Delta r_t$        | $\Delta r_t$       | $\Delta r_t$        | $\Delta r_t$        |
| $S_t$                 | 0.600***<br>(0.157)  | 0.545***<br>(0.151) | 0.433**<br>(0.211) | 0.427*<br>(0.232)   | 0.371*<br>(0.199)   |
| Inflation controls    |                      | 0.063**<br>(0.025)  | 0.060**<br>(0.024) | 0.087***<br>(0.026) | 0.048*<br>(0.025)   |
| Unemployment controls |                      | -0.033<br>(0.066)   | -0.020<br>(0.070)  | 0.045<br>(0.072)    | -0.020<br>(0.065)   |
| $S_t^p$               |                      |                     | 0.086<br>(0.069)   | 0.069<br>(0.076)    | 0.044<br>(0.079)    |
| $\Delta_t$            |                      |                     |                    | -0.148**<br>(0.062) | -0.116**<br>(0.056) |
| $\Delta r_{g,t-1}$    |                      |                     |                    |                     | 0.372***<br>(0.113) |
| Constant              | -0.085***<br>(0.031) | 0.047<br>(0.500)    | -0.035<br>(0.522)  | -0.580<br>(0.546)   | 0.014<br>(0.501)    |
| R-squared             | 0.602                | 0.629               | 0.633              | 0.652               | 0.704               |
| F-stat                | 14.596               | 7.233               | 8.158              | 6.533               | 7.879               |
| No. obs.              | 140                  | 140                 | 140                | 113                 | 113                 |

Standard errors in parentheses

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

Table 9: Weighted by dissent (0) or not (1) - majority "control"

|                       | (1)<br>$\Delta r_t$  | (2)<br>$\Delta r_t$ | (3)<br>$\Delta r_t$ | (4)<br>$\Delta r_t$ | (5)<br>$\Delta r_t$ |
|-----------------------|----------------------|---------------------|---------------------|---------------------|---------------------|
| $S_t$                 | 0.569***<br>(0.134)  | 0.523***<br>(0.128) | 0.483***<br>(0.169) | 0.533**<br>(0.206)  | 0.489***<br>(0.179) |
| Inflation controls    |                      | 0.060**<br>(0.024)  | 0.058**<br>(0.024)  | 0.082***<br>(0.026) | 0.038<br>(0.027)    |
| Unemployment controls |                      | -0.034<br>(0.064)   | -0.028<br>(0.067)   | 0.046<br>(0.094)    | -0.020<br>(0.077)   |
| $S_t^p$               |                      |                     | 0.035<br>(0.062)    | 0.010<br>(0.076)    | -0.008<br>(0.081)   |
| $\Delta_t$            |                      |                     |                     | -0.134<br>(0.095)   | -0.083<br>(0.088)   |
| $\Delta r_{g,t-1}$    |                      |                     |                     |                     | 0.383***<br>(0.114) |
| Constant              | -0.102***<br>(0.032) | 0.043<br>(0.481)    | 0.010<br>(0.500)    | -0.602<br>(0.712)   | 0.002<br>(0.600)    |
| R-squared             | 0.634                | 0.658               | 0.658               | 0.667               | 0.721               |
| F-stat                | 18.072               | 8.363               | 7.550               | 5.593               | 7.755               |
| No. obs.              | 140                  | 140                 | 140                 | 97                  | 97                  |

Standard errors in parentheses

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ 

Table 10: Keeping only chair of EB

|                       | (1)<br>$\Delta r_t$ | (2)<br>$\Delta r_t$ | (3)<br>$\Delta r_t$ |
|-----------------------|---------------------|---------------------|---------------------|
| $S_t$                 | 0.427***<br>(0.142) | 0.371***<br>(0.133) | 0.183<br>(0.116)    |
| Inflation controls    |                     | 0.084***<br>(0.025) | 0.075***<br>(0.022) |
| Unemployment controls |                     | -0.003<br>(0.097)   | 0.047<br>(0.091)    |
| $S_t^p$               |                     |                     | 0.217***<br>(0.067) |
| Constant              | -0.089**<br>(0.043) | -0.228<br>(0.730)   | -0.563<br>(0.685)   |
| R-squared             | 0.530               | 0.572               | 0.609               |
| F-stat                | 8.997               | 7.194               | 7.497               |
| No. obs.              | 107.000             | 107.000             | 107.000             |

Standard errors in parentheses

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ 

### A.8.2 Governor effect on interest rates

Now we will continue by looking at governor specific effects. We want to know if there are governors that are especially influential when it comes to the messaging conveyed in the press releases at the time of decisions. We also want to know if there are individual governors that deviate and by deviating from the others can

influence the decision in any direction. We will study this by first looking at the following regression:

$$S_t^p = \alpha_0 + \sum_{g \in G} \beta_g S_{gt} + \Gamma X_t + \varepsilon_t \quad (9)$$

$S_t^p$  is the sentiment of the preceding press release in time  $t$ ,  $S_{gt}$  is the sentiment score of governor  $g$  in time  $t$ . I expand the regression by including controls,  $X_t$  for inflation, unemployment and time fixed effects (year). We can see the results from this in Table 11. Almost all governors' statements are strongly correlated with the message that is conveyed in the preceding PR's. However the size of the correlation varies quite a lot. The big dissenters (LEO and KE) seem not to be less visible in the PR's.

Table 11: Results from regression (9)

|                                   | (1)<br>$S_t^p$      | (2)<br>$S_t^p$      |
|-----------------------------------|---------------------|---------------------|
| Aino Bunge $\times S_t$           | 0.696***<br>(0.242) | 0.761***<br>(0.242) |
| Anna Breman $\times S_t$          | 0.619***<br>(0.148) | 0.675***<br>(0.151) |
| Anna Seim $\times S_t$            | 0.547*<br>(0.300)   | 0.477<br>(0.319)    |
| Barbro Wickman-Parak $\times S_t$ | 0.676***<br>(0.095) | 0.699***<br>(0.097) |
| Cecilia Skingsley $\times S_t$    | 0.304***<br>(0.074) | 0.304***<br>(0.076) |
| Erik Thedéen $\times S_t$         | 0.921***<br>(0.296) | 1.047***<br>(0.306) |
| Henry Ohlsson $\times S_t$        | 0.262***<br>(0.056) | 0.310***<br>(0.066) |
| Irma Rosenberg $\times S_t$       | 0.774***<br>(0.153) | 0.955***<br>(0.280) |
| Karina Ekholm $\times S_t$        | 0.548***<br>(0.172) | 0.989***<br>(0.188) |
| Kerstin af Jochnick $\times S_t$  | 0.337***<br>(0.089) | 0.400***<br>(0.087) |
| Lars E.O. Svensson $\times S_t$   | 0.363***<br>(0.127) | 0.479***<br>(0.150) |
| Lars Nyberg $\times S_t$          | 0.728***<br>(0.118) | 0.917***<br>(0.099) |
| Martin Flodén $\times S_t$        | 0.421***<br>(0.082) | 0.407***<br>(0.076) |
| Per Jansson $\times S_t$          | 0.382***<br>(0.102) | 0.413***<br>(0.108) |
| Stefan Ingves $\times S_t$        | 0.498***<br>(0.064) | 0.604***<br>(0.074) |
| Svante Öberg $\times S_t$         | 0.695***<br>(0.093) | 0.812***<br>(0.088) |
| Inflation controls                | 0.092**<br>(0.043)  | 0.085*<br>(0.043)   |
| Unemployment controls             | -0.197*<br>(0.105)  | -0.200*<br>(0.101)  |
| Constant                          | 1.282<br>(0.851)    | 1.316<br>(0.827)    |
| Governor FE                       | No                  | Yes                 |
| R-squared                         | 0.719               | 0.739               |
| F-stat                            | 8                   | 11                  |
| No. obs.                          | 624                 | 624                 |

Standard errors in parentheses

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

## A.9 Variance analysis

To further analyze whether the results on reduced size effects on short term OIS rates, we look to data on volatility and difference between the highest and lowest quote price around the releases of .

$$var(Y_t) = \alpha_0 + \beta_0 \tilde{d}_t + \varepsilon_t \quad (10)$$

Table 12: Results from regression 10

|                   | (1)<br>1M         | (2)<br>1M         | (3)<br>3M         | (4)<br>3M         | (5)<br>OMXSPI     | (6)<br>OMXSPI     |
|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| $\Delta_t$        | 0.000<br>(0.000)  | 0.000<br>(0.000)  | 0.001<br>(0.001)  | 0.001<br>(0.001)  | 0.087<br>(0.142)  | 0.042<br>(0.129)  |
| $r_t$             | 0.000<br>(0.000)  | 0.000<br>(0.000)  | 0.001<br>(0.001)  | 0.001<br>(0.001)  | 0.272<br>(0.241)  | 0.017<br>(0.100)  |
| VIX               |                   | 0.000<br>(0.000)  |                   | 0.001<br>(0.000)  |                   | -0.027<br>(0.061) |
| Constant          | -0.000<br>(0.000) | -0.000<br>(0.000) | -0.001<br>(0.001) | -0.001<br>(0.001) | -0.399<br>(0.285) | -0.085<br>(0.096) |
| Year FE           | Yes               | Yes               | Yes               | Yes               | Yes               | Yes               |
| R-squared         | 0.523             | 0.524             | 0.236             | 0.372             | 0.173             | 0.059             |
| Mean outcome size |                   |                   |                   |                   |                   |                   |
| F-stat            | 1                 | 1                 | 0                 | 1                 | 1                 | 0                 |
| No. obs.          | 39                | 39                | 56                | 56                | 109               | 108               |

Standard errors in parentheses

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

$$maxmin(Y_t) = \alpha_0 + \beta_0 \tilde{d}_t + \varepsilon_t \quad (11)$$

Table 13: Results from regression 11

|                   | (1)               | (2)               | (3)               | (4)               | (5)               | (6)               |
|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
|                   | 1M                | 1M                | 3M                | 3M                | OMXSPI            | OMXSPI            |
| $\Delta_t$        | 0.006<br>(0.009)  | 0.009<br>(0.010)  | 0.055<br>(0.049)  | 0.031<br>(0.031)  | 0.079<br>(0.253)  | 0.005<br>(0.245)  |
| $r_t$             | 0.021<br>(0.019)  | 0.046<br>(0.034)  | 0.043<br>(0.034)  | 0.052<br>(0.033)  | 0.217<br>(0.320)  | -0.008<br>(0.262) |
| VIX               |                   | 0.004<br>(0.008)  |                   | 0.031<br>(0.020)  |                   | 0.066<br>(0.122)  |
| Constant          | -0.013<br>(0.019) | -0.041<br>(0.037) | -0.035<br>(0.042) | -0.055<br>(0.046) | -0.495<br>(0.366) | -0.218<br>(0.281) |
| Year FE           | Yes               | Yes               | Yes               | Yes               | Yes               | Yes               |
| R-squared         | 0.361             | 0.415             | 0.260             | 0.388             | 0.136             | 0.112             |
| Mean outcome size |                   |                   |                   |                   |                   |                   |
| F-stat            | 1                 | 1                 | 1                 | 1                 | 0                 | 0                 |
| No. obs.          | 61                | 59                | 70                | 68                | 109               | 108               |

Standard errors in parentheses

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ 

## A.10 Robustness checks of head line result

Table 14: Robustness check of m1 result looking at sign restricted shocks

|                                          | (1)                 | (2)               |
|------------------------------------------|---------------------|-------------------|
|                                          | 3m                  | 3m                |
| $\text{lm}_{dissent_d} \text{direction}$ | -0.908<br>(0.544)   | 0.072<br>(0.289)  |
| Constant                                 | 0.962***<br>(0.353) | -0.210<br>(0.203) |
| R-squared                                | 0.253               | 0.199             |
| F-stat                                   | 2.786               | 0.063             |
| No. obs.                                 | 61.000              | 73.000            |

Standard errors in parentheses

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ 

Table 15: Robustness check of m1 result

|                                          | (1)                 | (2)                 | (3)                 | (4)                 | (5)                 | (6)                 |
|------------------------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                                          | 3m                  | 3m                  | 3m                  | 3m                  | 3m                  | 3m                  |
| $\text{lm}_{dissent_d} \text{direction}$ | -0.500<br>(0.323)   | -0.615*<br>(0.360)  | -0.225<br>(0.200)   | -1.349**<br>(0.539) | -0.346<br>(0.470)   | -0.791*<br>(0.402)  |
| Constant                                 | 1.158***<br>(0.148) | 1.178***<br>(0.200) | 1.166***<br>(0.179) | 1.080**<br>(0.449)  | 1.026***<br>(0.372) | 1.460***<br>(0.226) |
| Sample                                   | Drop LEO            | Drop t/b 5 %        | No vote dissent     | With vote dissent   | 2007-2015           | 2015-2025           |
| R-squared                                | 0.297               | 0.199               | 0.224               | 0.315               | 0.132               | 0.316               |
| Mean outcome size                        |                     |                     |                     |                     |                     |                     |
| F-stat                                   | 2.388               | 2.926               | 1.270               | 6.268               | 0.542               | 3.873               |
| No. obs.                                 | 106.000             | 104.000             | 63.000              | 36.000              | 45.000              | 64.000              |

Standard errors in parentheses

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

## References

- Almerud, J., D. Krygier, H. Lundvall, and M. Njie (Dec. 2024). “Measuring Riksbank Monetary Policy: Shocks and Macroeconomic Transmission”. In: *Sveriges Riksbank Working Paper Series* 445. URL: <https://www.riksbank.se/globalassets/media/rapporter/working-papers/2024/no.-445-measuring-riksbank-monetary-policy-shocks-and-macroeconomic-transmission.pdf>.
- Apel, M. and M. Blix Grimaldi (Apr. 2012). “The Information Content of Central Bank Minutes”. In: *Riksbank Research Paper Series* 92. URL: <https://ssrn.com/abstract=2092575>.
- (Apr. 2014). “How Informative Are Central Bank Minutes?” In: *Review of Economics* 65.1, pp. 53–76. DOI: 10.1515/roe-2014-0104. URL: <https://ideas.repec.org/a/lus/reveco/v65y2014i1p53-76.html>.
- Apel, M., M. Blix Grimaldi, L. Ahrenberg, and A. Jönsson (Feb. 2026). *The effect of increased transparency on an individualistic monetary policy committee*. Working Paper Series 463. Sveriges Riksbank (Central Bank of Sweden). DOI: None. URL: <https://ideas.repec.org/p/hhs/rbnkwp/0463.html>.
- Apel, M., M. Blix Grimaldi, and I. Hull (2022). “How Much Information Do Monetary Policy Committees Disclose? Evidence from the FOMC’s Minutes and Transcripts”. In: *Journal of Money, Credit and Banking* 54.5, pp. 1459–1490. DOI: <https://doi.org/10.1111/jmcb.12885>. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/jmcb.12885>. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1111/jmcb.12885>.
- Baird, C., J. Benchimol, V. Vyshnevskaya, W. Sohn, and I. Vyshnevskiy (None 2026). *The monetary policy statement database*. DOI: None. URL: <https://ideas.repec.org/h/bis/bisifc/67-14.html>.
- Blinker, A. S., M. Ehrmann, M. Fratzscher, J. De Haan, and D.-J. Jansen (Dec. 2008). “Central Bank Communication and Monetary Policy: A Survey of Theory and Evidence”. In: *Journal of Economic Literature* 46.4, pp. 910–45. DOI: 10.1257/jel.46.4.910. URL: <https://www.aeaweb.org/articles?id=10.1257/jel.46.4.910>.
- Blot, C., P. Hubert, and F. Labondance (2025). *Dissent in Monetary Policy Decisions: Effects, Channels and Implications*. Working Papers hal-05158109. HAL. DOI: None. URL: <https://ideas.repec.org/p/hal/wpaper/hal-05158109.html>.
- Fiore, F. D., A. Maurin, A. Mijakovic, and D. Sandri (Dec. 2024). *Monetary policy in the news: communication pass-through and inflation expectations*. BIS Working Papers 1231. Bank for International Settlements. DOI: None. URL: <https://ideas.repec.org/p/bis/biswps/1231.html>.
- Geiger, F., D. Kanelis, P. Lieberknecht, and D. Sola (2025). *Monetary-Intelligent Language Agent (MILA)*. Technical Papers 01/2025. Deutsche Bundesbank. DOI: None. URL: <https://ideas.repec.org/p/zbw/bubtps/316448.html>.
- Grattafiori, A. et al. (2024). *The Llama 3 Herd of Models*. arXiv: 2407.21783 [cs.AI]. URL: <https://arxiv.org/abs/2407.21783>.

- Hansen, S., M. McMahon, and A. Prat (None 2018). “Transparency and Deliberation Within the FOMC: A Computational Linguistics Approach”. In: *The Quarterly Journal of Economics* 133.2, pp. 801–870. DOI: [None](#). URL: <https://ideas.repec.org/a/oup/qjecon/v133y2018i2p801-870..html>.
- Howes, C., M. Dordal i Carreras, O. Coibion, and Y. Gorodnichenko (Jan. 2026). *How Monetary Policy Is Made: Lessons from Historical FOMC Discussions*. Working Paper 34632. National Bureau of Economic Research. DOI: [10.3386/w34632](#). URL: <http://www.nber.org/papers/w34632>.
- Jordà, Ò. (Mar. 2005). “Estimation and Inference of Impulse Responses by Local Projections”. In: *American Economic Review* 95.1, pp. 161–182. DOI: [10.1257/0002828053828518](#). URL: <https://www.aeaweb.org/articles?id=10.1257/0002828053828518>.
- Meade, E. and D. Stasavage (Apr. 2008). “Publicity of Debate and the Incentive to Dissent: Evidence from the US Federal Reserve”. In: *Economic Journal* 118.528, pp. 695–717. DOI: [None](#). URL: <https://ideas.repec.org/a/ecj/econjl/v118y2008i528p695-717.html>.
- Meta AI (2025). *Llama 4 Multimodal Intelligence*. <https://ai.meta.com/blog/llama-4-multimodal-intelligence/>. Accessed: 2026-01-08. Meta Platforms, Inc. URL: <https://ai.meta.com/blog/llama-4-multimodal-intelligence/>.
- Riboni, A. and F. Ruge-Murcia (2014). “Dissent in monetary policy decisions”. In: *Journal of Monetary Economics* 66, pp. 137–154. ISSN: 0304-3932. DOI: <https://doi.org/10.1016/j.jmoneco.2014.03.006>. URL: <https://www.sciencedirect.com/science/article/pii/S030439321400049X>.
- Sveriges Riksbank (2009-04-02). “Protokollsbilaga A, Direktionens protokoll 090402, § 4”. Avdelning: STA/Informationssekr

## A.11 Robustness, dovishness result

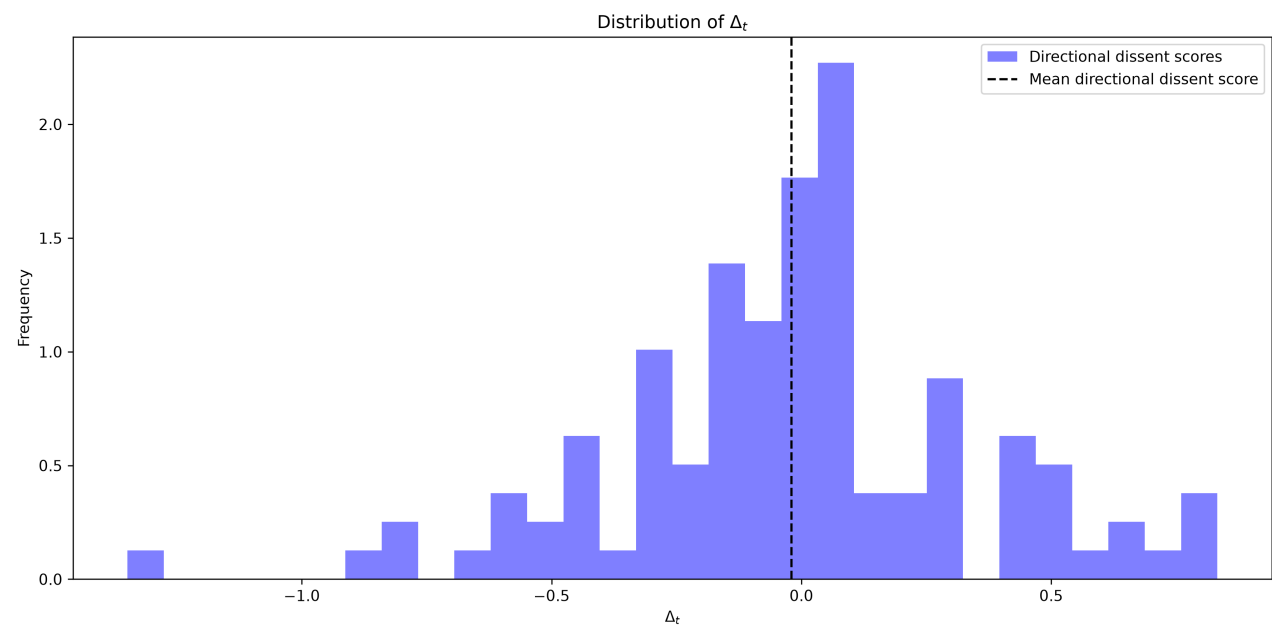
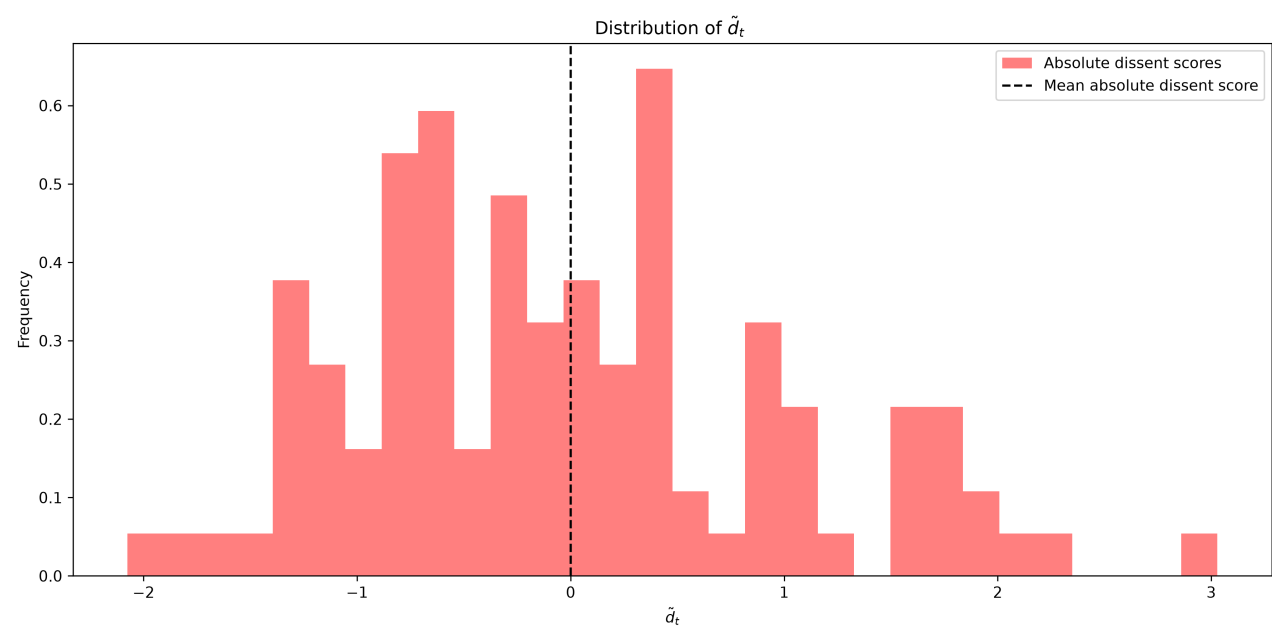
|                   | (1)               | (2)                 | (3)                 | (4)                 | (5)               | (6)                 | (7)               | (8)                 | (9)                 | (10)               |
|-------------------|-------------------|---------------------|---------------------|---------------------|-------------------|---------------------|-------------------|---------------------|---------------------|--------------------|
|                   | 1M                | 3M                  | 6M                  | 1Y                  | 2Y                | 5Y                  | 10Y               | SEK/USD             | SEK/EUR             | OMX                |
| $\Delta_t$        | 0.252<br>(0.575)  | -1.358**<br>(0.653) | -0.308<br>(0.432)   | -0.317*<br>(0.158)  | -0.321<br>(0.202) | -0.179<br>(0.174)   | -0.185<br>(0.199) | 4.963<br>(2.981)    | 5.410*<br>(2.772)   | 0.883<br>(2.849)   |
| Majority mean     | 2.030<br>(3.138)  | -1.170<br>(2.883)   | -0.887<br>(1.580)   | -0.867<br>(0.838)   | -1.189<br>(0.894) | -0.298<br>(0.683)   | -0.425<br>(0.708) | 12.298<br>(12.766)  | 13.754<br>(10.537)  | -9.922<br>(21.712) |
| Dissent votes     | -0.751<br>(0.899) | -0.153<br>(0.507)   | 0.174<br>(0.491)    | 0.478<br>(0.331)    | 0.090<br>(0.337)  | -0.064<br>(0.200)   | 0.042<br>(0.199)  | 2.464<br>(4.120)    | -1.028<br>(2.647)   | -10.218<br>(8.006) |
| $r_t$             | 2.426<br>(1.971)  | 0.948<br>(1.148)    | -1.777**<br>(0.784) | -0.437<br>(0.501)   | -0.718<br>(0.578) | -1.004**<br>(0.383) | -0.576<br>(0.384) | 9.890<br>(7.120)    | 11.591*<br>(6.307)  | 4.919<br>(13.923)  |
| VIX               | 0.980<br>(1.011)  | -0.055<br>(0.751)   | 0.290<br>(0.821)    | -0.727**<br>(0.280) | 0.250<br>(0.442)  | -0.007<br>(0.292)   | 0.193<br>(0.284)  | 5.193<br>(3.900)    | 6.525*<br>(3.341)   | -8.153<br>(5.940)  |
| Constant          | -3.909<br>(3.116) | -1.946<br>(1.704)   | 3.166**<br>(1.193)  | -0.134<br>(0.697)   | 0.771<br>(0.854)  | 1.240*<br>(0.621)   | 0.585<br>(0.685)  | -16.914<br>(11.392) | -17.272<br>(10.461) | -5.654<br>(21.032) |
| Year FE           | Yes               | Yes                 | Yes                 | Yes                 | Yes               | Yes                 | Yes               | Yes                 | Yes                 | Yes                |
| R-squared         | 0.340             | 0.503               | 0.376               | 0.516               | 0.281             | 0.302               | 0.235             | 0.326               | 0.298               | 0.458              |
| Mean outcome size | -.134             | .192                | .338                | -.383               | -.454             | -.429               | -.342             | -.066               | .114                | -6.994             |
| F-stat            | 0.386             | 1.673               | 1.193               | 3.161               | 1.290             | 2.167               | 1.916             | 1.029               | 1.534               | 0.890              |
| No. obs.          | 48.000            | 48.000              | 48.000              | 48.000              | 48.000            | 48.000              | 48.000            | 48.000              | 48.000              | 48.000             |

Standard errors in parentheses

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

Table 16: Result robust to extraordinarily deviating governor

A.12 Distributions



Mean negative independent variable size (-0.26)